Arista Access Node QoS/Congestion Event Troubleshooting and Resolution SOP

Node: KR-ACC-20 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_9889

Description

This Standard Operating Procedure (SOP) is designed to address QoS/Congestion Events, specifically microbursts, interface tail drops, and QoS queue starvation, on Arista access nodes, such as KR-ACC-20 located at KR Puram Access. These events occur due to transient periods of high network utilization, often caused by sudden changes in traffic patterns, which can overwhelm the buffering capabilities of network interfaces, leading to packet drops and degradation of network performance. The Arista access node, operating in a role that aggregates traffic from various sources, is particularly susceptible to these events due to its position in the network hierarchy. Understanding the underlying mechanics of QoS policies, queue management, and scheduling algorithms on Arista equipment is crucial for effective troubleshooting and resolution. The EOS (Arista Extensible Operating System) provides a robust set of tools and commands for diagnosing and mitigating these issues, leveraging its advanced traffic management capabilities.

Pre-requisite Checks

- Verify KR Puram Access site environmental conditions are within operational norms
- Confirm redundant link status to ensure no single points of failure
- Dispatch remote hands to KR Puram Access for potential hardware interventions
- Review recent network configuration changes for potential impact

Potential Root Causes

Cause	Probability
Inadequate QoS Policy Configuration	High
Insufficient Bandwidth Allocation	Medium
Hardware Failure or Degradation	Low

Diagnostic Commands & Expected Outputs

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Command: show qos interface eth1, Expected Output: Displays the current QoS configuration and statistics for the specified interface, Description: This command isolates the QoS configuration applied to the interface, helping identify if the current policy is adequate for the traffic profile or if adjustments are needed to prevent congestion

Command: show interface eth1 counters, Expected Output: Provides detailed interface counters, including drops and errors, Description: Analyzing interface counters can help pinpoint the exact nature of the
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congestion event, such as tail drops or queue overflows

Command: show running-config section qos, Expected Output: Displays the current running QoS configuration, Description: Reviewing the QoS configuration section of the running config helps in understanding how traffic is currently being managed and if there are any misconfigurations
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Resolution Steps

Step 1: Step: Adjust QoS Policy, Command: qos policy-map QOS_POLICY_1 class CLASS_1 police rate 100mb burst 100k, Description: Adjusting the QoS policy to more accurately reflect the traffic requirements and prevent congestion, this involves modifying the police rate and burst size to better handle transient traffic spikes

Step 2: Step: Increase Bandwidth Allocation, Command: interface eth1 bandwidth 1000mb, Description: Increasing the bandwidth allocation on the interface can help mitigate congestion by providing more capacity for traffic, though this should be done judiciously to avoid oversubscription

Step 3: Step: Apply QoS Queue Management, Command: qos queue-map QOS_QUEUE_MAP_1 queue 1 percent 50, Description: Configuring QoS queue management ensures that critical traffic is prioritized and less critical traffic is appropriately limited, preventing starvation of critical queues

Post-Check Verification

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Command: show qos interface eth1 statistics, Expected Output: Displays updated QoS statistics for the interface, reflecting the changes made, Description: Verifying the QoS interface statistics after implementing changes helps confirm if the adjustments have successfully mitigated the congestion event

Command: show interface eth1 counters detail, Expected Output: Provides detailed, updated interface counters, Description: Reviewing the detailed interface counters after changes are applied can confirm if drop rates have decreased and if the interface is operating within expected parameters
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Roll-back Procedures

- Step: Revert QoS Policy, Command: no qos policy-map QOS_POLICY_1, Description: If the adjustments made to the QoS policy do not resolve the issue or introduce new problems, reverting to the previous policy ensures that the network returns to a known state

- Step: Restore Original Bandwidth Allocation, Command: no interface eth1 bandwidth 1000mb, Description: If increasing bandwidth allocation does not mitigate congestion or causes other issues, restoring the original bandwidth allocation is necessary to avoid unintended network behavior